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Palaeomagnetism and Potassium-Argon Ages of Volcanic Rocks of Ngorongoro Caldera, Tanzania*

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Summary

Natural remanent magnetization (NRM) has been measured in 21 lava flows and 1 tuff in the south-west wall of Ngorongoro caldera, Tanzania. The lowest three lavas are normally magnetized, the next two have intermediate directions, and the remainder are reversed; potassium–argon dating places the reversal at 2·45 My, corresponding to the Gauss–Matuyama epoch boundary. Alternating-field demagnetization was effective in removing the ubiquitous lightning-produced secondary magnetizations, but partial thermal demagnetization was not. Curie temperatures of the lavas fell into two groups: 250°–350 °C and 570°–585 °C, probably corresponding to unoxidized and oxidized titanomagnetites respectively. NRM directions in the lavas occurred in several distinct and stratigraphically continuous groups, suggesting intermittent eruption of groups of lavas. No correspondence was found between NRM groups and other parameters, such as Curie temperatures. The dispersion of mean NRM directions is large in comparison with late Quaternary lavas from low latitudes in the Pacific basin, but probably is not large relative to the proportion of the geomagnetic field in Africa that is non-dipolar. The mean palaeomagnetic pole is offset 10·5° from the geographic pole, but because of the large dispersion, the difference is not considered significant.

Introduction

A section more than 340 m thick of late Cenozoic volcanic rocks forms the walls of the sixteen-mile-wide caldera of Ngorongoro, northern Tanzania. The rocks are predominantly trachytic lavas† with subordinate olivine basalts and interbedded tuffs and agglomerates. A preliminary investigation in 1964 by one of us (R. L. Hay) using a portable fluxgate magnetometer (Doell & Cox 1962) showed that a geomagnetic transition from normal to reversed polarity was recorded in the lavas. Because some of the lavas at the polarity transition seemed to have intermediate directions of remanent magnetization, a small number of oriented hand specimens was collected in that part of the section. The results of magnetic measurements of these samples indicated that more thorough sampling was warranted; this was done

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† Most of the lavas herein called trachytes are strictly speaking trachyandesites, but our petrographic examination was not sufficiently detailed to distinguish the two types.

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in 1967 with a portable diamond core drill. The major objectives of this work have been (1) to date the polarity reversal, (2) to investigate the suitability of the lavas for determining the palaeointensity of the geomagnetic field, particularly in the region of the polarity transition, (3) to seek possible correlations of magnetic and petrologic characteristics, and (4) to compare the effectiveness of partial a.f. demagnetization and partial thermal demagnetization in removing secondary components of natural remanent magnetization.

At least 36 separate lavas, together with tuffs and agglomerates, are exposed along the road leading down the south-west wall of the caldera to Lerai Camp, but not all the lavas are sufficiently well exposed for adequate palaeomagnetic sampling. Reported here are magnetic measurements for an average of five specimens from 23 lava flows and one welded tuff and potassium–argon age determinations on one of the lavas and an additional tuff. An index map of Ngorongoro caldera and the locations of the sampling sites are shown in Fig. 1, and the stratigraphic distribution of the various volcanic rock types and the lavas sampled are shown in Fig. 2.

Remanent magnetization

The measurements of natural remanent magnetization (NRM) after partial alternating field demagnetization are summarized in Table 1 and Fig. 3. All of the rocks in the section were found to carry very large components of secondary magnetization. Alternating fields of 200 peak Oe were generally required to remove these secondary magnetizations sufficiently to provide reasonable groupings of NRM directions within single flows. The NRM's prior to demagnetization were highly variable in intensity as well as direction, an aspect characteristic of extensive remagnetization by lightning.

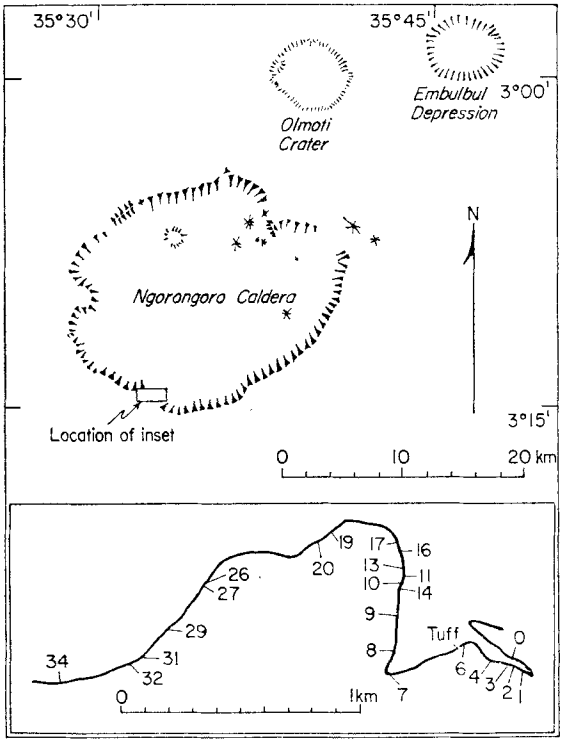


FIG. 1. Sketch map of Ngorongoro caldera, northern Tanzania. Inset shows locations of volcanic units sampled along the Lerai Road.

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Table 1

Summary of stratigraphic position, magnetic characteristics, and lithology of all volcanic units sampled

Flow number	Relative elevation (m)	Polarity	N	D	I	R	k	α_{95}	Demagnetizing field (oersteds)	k_0	$J_S T$ type	Rock type	NRM group
34	315	R	5	193·1	6·6	4·83037	23·6	16·1	200	5·5	1	T(F)	
32	285	R	6	177·5	1·8	5·98378	308·0	3·8	200	41·0	1a	T(F, X)	
31	278	R	6	182·1	24·7	5·67627	15·5	17·6	100	1·8	1a	T(F, X)	
29	261	R	6	179·9	35·6	5·90162	50·8	9·5	200	1·4	1a, 2	T(F)	E
27	242	R	7	166·3	4·6	6·96291	162·0	4·8	200	2·3	3	OB(P)	
26	238	R	5	165·4	6·9	4·71820	14·2	21·0	100	1·1	1	OB(V, P)	
20	181	R	8	172·3	10·7	7·26590	9·5	18·9	200	1·8	1	T(F, P)	
19	176	R	5	181·8	7·4	4·98264	230·0	5·1	200	1·0	1	T(V, P, O)	
17	141	R	3	154·1	−30·7	2·98209	112·0	11·7	200	7·3	1, 2	T(P)	
16	136	R	6	176·9	−23·3	5·95893	122·0	6·1	200	1·8	2	T(V, P)	D
14	119	R	5	173·3	−33·0	4·99332	598·0	3·1	200	34·5	1a	T(P)	
13	128	R	6	219·3	−27·3	5·90993	55·5	9·1	200	2·7	1a, 2	T(V, P)	
11	124	R	4	202·3	−26·7	3·89090	27·5	17·8	200	2·3	2	T(V, P)	
10	121	R	4	202·5	−38·2	3·99557	677·0	3·5	200	23·2	1, 2	T(V, P)	
9	106	R?	4	—	—	2·14680	1·6	—	(200)	1·1	1a	T(P)	C
8	92	R	4	204·4	−53·3	3·98901	273·0	5·6	200	1·7	2	T(P)	
7	80	R	4	216·9	−38·5	3·88982	27·2	17·9	50	3·8	2	T(P)	
Tuff	46	R	4	166·9	1·5	3·91513	35·4	15·7	100	1·7	1a	Tuff	
6	45	R?	4	—	—	2·19137	1·7	—	(400)	1·5	1	T(P, O)	
4	31	I	5	65·6	41·6	4·91918	49·5	11·0	200	8·3	—	T(P)	
3	24	I	5	308·9	2·1	4·96379	110·0	7·3	200	36·9	1	T(V, P, O)	B
2	20	N	5	25·0	0·7	4·65514	11·6	23·5	200	0·9	1	T(V, P, O)	
1	16	N	4	353·0	7·6	3·97233	108·0	8·9	200	1·8	1	T(V, P, O)	A
0	0	N	5	4·4	−4·0	4·92431	52·9	10·6	200	11·3	—	T(P)	

Explanation: N, number of samples. D, declination east of north. I, inclination, positive downward. R, k, α_{95} (Fisher 1953) statistical parameters after a.f. demagnetization. k_0, precision parameter before a.f. demagnetization. Rock type symbols: T, trachyte. OB, olivine basalt. F, fine grained. X, contains xenolithic fragments. P, porphyritic. V, vesicular. O, oxidized. Polarity: R, reversed; I, intermediate; N, normal. See text for explanation of $J_S - T$ type.

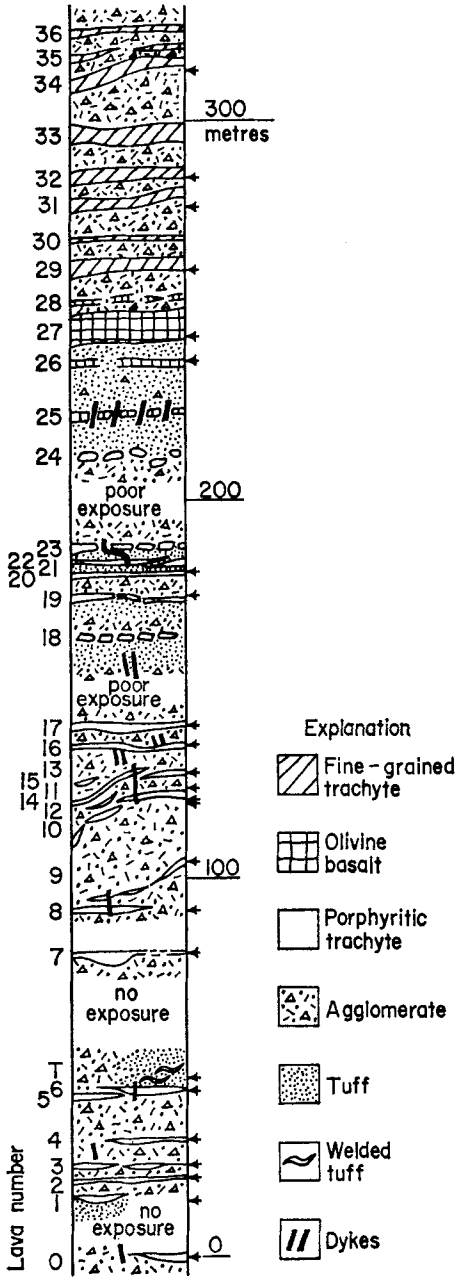


FIG. 2. Diagrammatic geologic section of the wall of Ngorongoro caldera constructed from exposures along the Lerai Road. Positions of volcanic units sampled for palaeomagnetic study are indicated by arrows.

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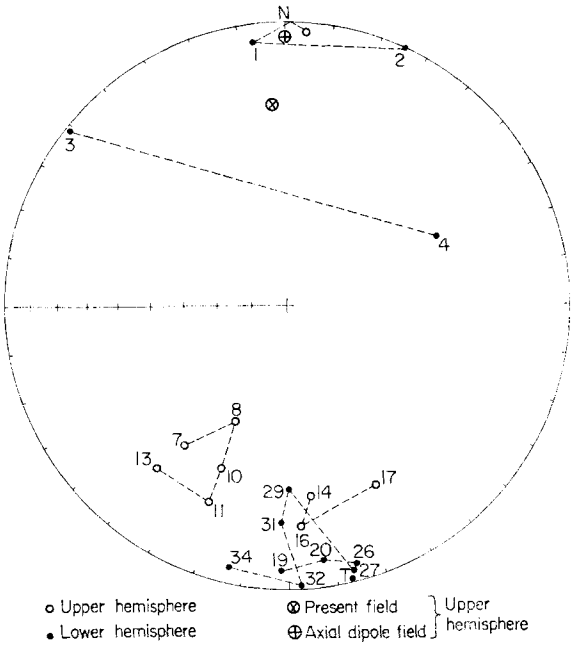


FIG. 3. Mean north-seeking NRM directions in 22 volcanic units after a.f. demagnetization. Numbers represent lava flows as shown in Fig. 2; '*T*' is the welded tuff stratigraphically between flows 4 and 7. Missing numbers represent lava flows that either could not be magnetically cleaned or were not sampled. Dashed lines connect directions within NRM groups (see text). Equal-area projection, circumference is horizontal plane.

Prior to demagnetization, 16 of the 24 sets of samples measured were found to be randomly magnetized; nevertheless all but two of the lavas were effectively 'cleaned'. Comparison of the values of k_0 (the precision parameter for the initial measurements) with k (the precision parameter after a.f. demagnetization) in Table 1 shows that for most of the rocks, directions were much better grouped after demagnetization. Nevertheless, the radii (α_{95}) of the 95-per-cent-confidence circles after demagnetization range upward to 21° (Table 1). This is rather low precision for such young lavas (see, e.g., Doell & Cox 1965), and the reason seems to be that the TRM's of these rocks have comparatively low coercivity spectra (Larson *et al.* 1969). A few samples were so completely remagnetized that the presumed original NRM direction could not be recovered; of the 22 sites that were effectively 'cleaned', 10 had one or two such samples. NRM directions in these discarded samples were at least 45° from the mean of the remainder, and the discrepancy was usually of the order of 90°. A.F. demagnetization could not be carried beyond 200 peak Oe because above this value the spurious magnetizations produced by the demagnetizing apparatus (Doell & Cox 1967a) became comparable in magnitude to the remaining NRM.

In Fig. 3 it can be seen that the directions of NRM in the various lavas are distinctly grouped, and that each group can also be considered a stratigraphic unit. The grouping is as follows: (A) lavas 0, 1 and 2; (B) lavas 3 and 4; (C) lavas 7, 8, 10, 11, and 13; (D) lavas 14, 16 and 17; (E) lavas 19, 20, 26, 27, 31, 32 and 34. Note that although the welded tuff seems to belong magnetically to group E, its stratigraphic position is between groups B and C. The transition from normal to reversed polarity occurs between lava 2 (normal) and the welded tuff (reversed), and although the NRM's of lavas 3 and 4 are divergent, they are placed in a single group because

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they are clearly intermediate between normal and reversed. (An intermediate NRM direction is conveniently defined as one that corresponds to a virtual geomagnetic pole with latitude less than 45°.) There is some indication of serial correlation of directions (Doell 1969) in the groups in which the number of lavas is sufficient for this to be evident, i.e. groups C and E. Application of the test of Watson & Beran (1967) to these two groups, however, shows no significant serial correlation at the 95-per-cent probability level. Hence we conclude that the lavas within groups C and E were not erupted sufficiently rapidly to have recorded geomagnetic secular variation quasi-continuously; in contrast to the results found by Doell (1969) for a sequence of Hawaiian lavas. In any event, the grouping of NRM directions strongly suggests that the lavas of Ngorongoro were erupted intermittently in groups.

Thermomagnetic properties

Because these lavas span a polarity transition and two of them clearly have intermediate NRM directions, it seemed desirable to use them to attempt determinations of geomagnetic palaeointensity. Large secondary magnetizations diminish the likelihood of success in this, so thermal demagnetization trials were made to see if the secondary magnetization could be removed at relatively low temperatures. Thermal demagnetization was accomplished by heating and cooling the samples in an atmosphere of dry oxygen-free nitrogen in a field less than 100 γ and measuring the remaining NRM at room temperature. The experiments were done on the preliminary collection of hand samples. To aid in the interpretation of the results, thermomagnetic curves in fields ranging from 1500 to 3000 Oe were obtained on fresh specimens of the same samples, also in nitrogen atmosphere, using the automatic recording balance described by Doell & Cox (1967b). The thermomagnetic results are given in Figs 4–8, in which are shown the changes in NRM directions after progressive heatings, the mean direction of NRM in the lavas obtained by a.f. demagnetization, and the variation of NRM and strong-field magnetization with temperature.

Large secondary components of magnetization are seen to be present in nearly all samples, as for example, in that from lava 13 (Fig. 4). On successive heatings the NRM direction moved closer to the mean of the a.f. demagnetized samples. However, it did not move within the 95-per-cent confidence circle about the mean, a characteristic common to all our thermal demagnetization experiments. The results from lavas 10, 11, 16, 17 and 34 (Figs 5, 6 and 8) are similar in the sense that after heating, the specimen directions move toward the mean direction but do not reach it. The only specimen that did not have a large secondary component was that from lava 29 (Fig. 7); here the NRM direction remained close to the mean after heating to temperatures as high as 500 °C.

In the samples from lavas 1 (Fig. 4) and 27 (Fig. 7) the NRM directions clearly moved away from the stable mean direction on heating. A possible explanation for this apparent tendency for 'self-reversals' is that samples became magnetized in the fields of adjacent samples during cooling. In our experiments the specimens were heated together in one oven and were separated by distances of about half their diameters. Although the intensities of NRM were not especially great (3×10^{-3} to 1×10^{-4} emu cm^{-1} cc), the diversity of Curie temperatures allowed the possibility that cooling in the fields of adjacent specimens might produce spurious TRM in some specimens with low Curie temperatures. Therefore, midway in these experiments some specimens (from lavas 1, 10, 17 and 27, which showed the most erratic behaviour) were reheated to temperatures 2° or 3 °C higher than the previous heating and cooled individually. Results of this test were not systematic: in one case, only the direction changes; in another, only the intensity; and in the other two, both

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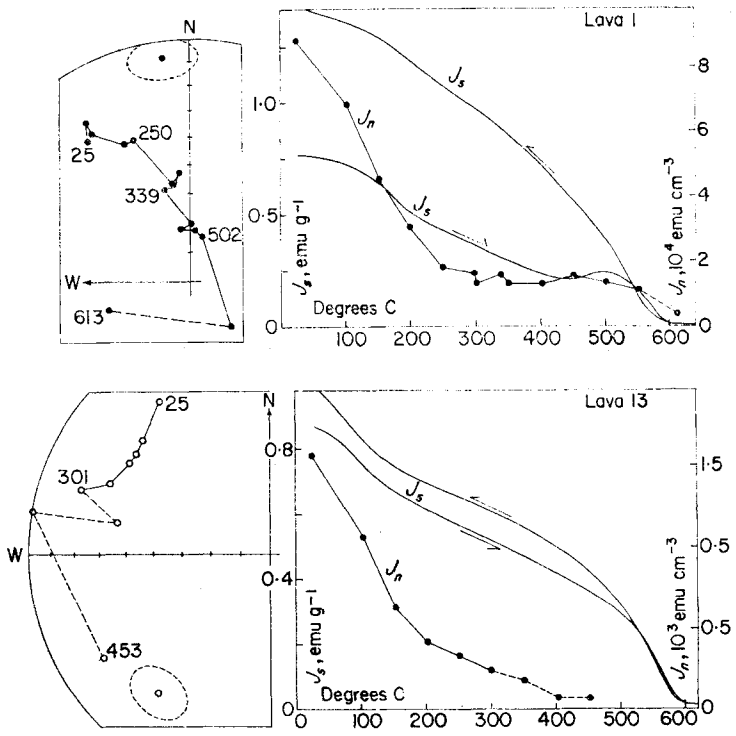


FIG. 4. Results of heating experiments on samples from lavas 1 and 13. On the segments of equal-area projections (left), the mean NRM directions after a.f. demagnetization of nearly all samples from each flow, enclosed by the dashed 95-per-cent-confidence circles; the directions of the NRM in one sample from each flow, at room temperature (25) and after heating to the temperatures indicated and cooling in nearly zero field. Solid circles indicate positive inclinations, open circles negative inclinations. J_n, the intensities of NRM in these samples after each heating step; dashes indicate where the NRM was unstable or partially stable. J_s, the heating and cooling curves for strong-field magnetization.

direction and intensity changed. The proximity of specimens during cooling does not seem to explain the apparent ' self-reversal ' tendencies of lavas 1 and 27, although it may account in part for irregularities in demagnetization curves.

Because of the ineffectiveness of thermal demagnetization in removing secondary NRM components, no attempts have been made to obtain palaeomagnetic intensities from the Ngorongoro lavas. A preliminary attempt by Coe (1967) to combine a.f. demagnetization with the Thellier & Thellier (1959) stepwise heating method in order to determine palaeointensities gave largely negative results, but experiments by Doell & Smith (1969) indicate that in some instances a.f. demagnetization may be successfully combined with stepwise heatings. Thus the possibility remains that valid determinations of palaeointensity could be made using rocks such as the Ngorongoro lavas. It is worth noting also that in contrast to our results for lightning-produced secondary magnetizations, Wilson & Smith (1968) found partial thermal demagnetization effective in removing secondary magnetization of viscous origin from a variety of lavas.

The strong-field magnetization curves shown in Figs 4–8 indicated that there is a possibility of grouping the lavas according to magneto-mineralogical type. Accordingly, strong-field thermomagnetic curves were obtained in air for specimens of

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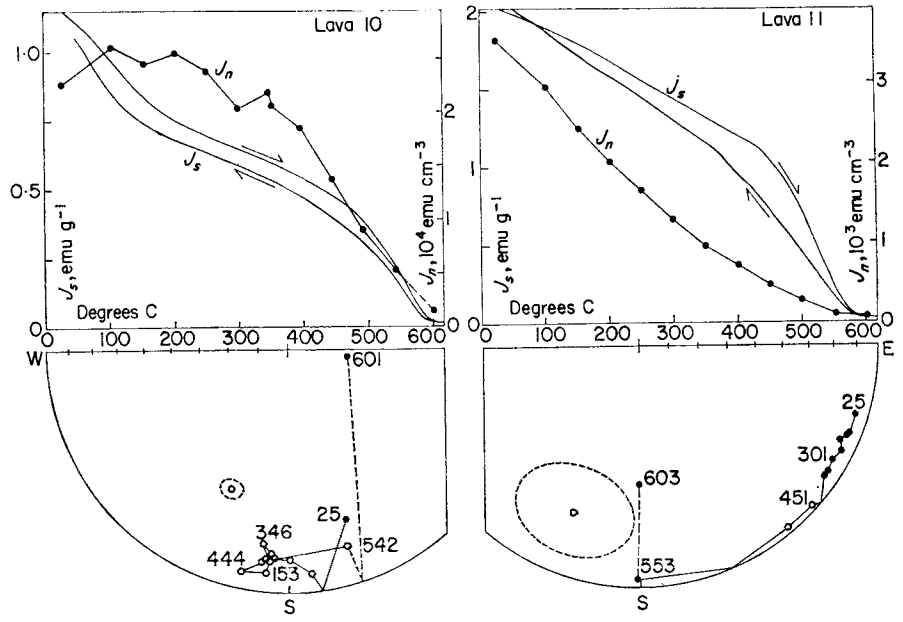


FIG. 5. Thermal demagnetization of NRM and strong-field magnetization, lavas 10 and 11. See Fig. 4 for explanation.

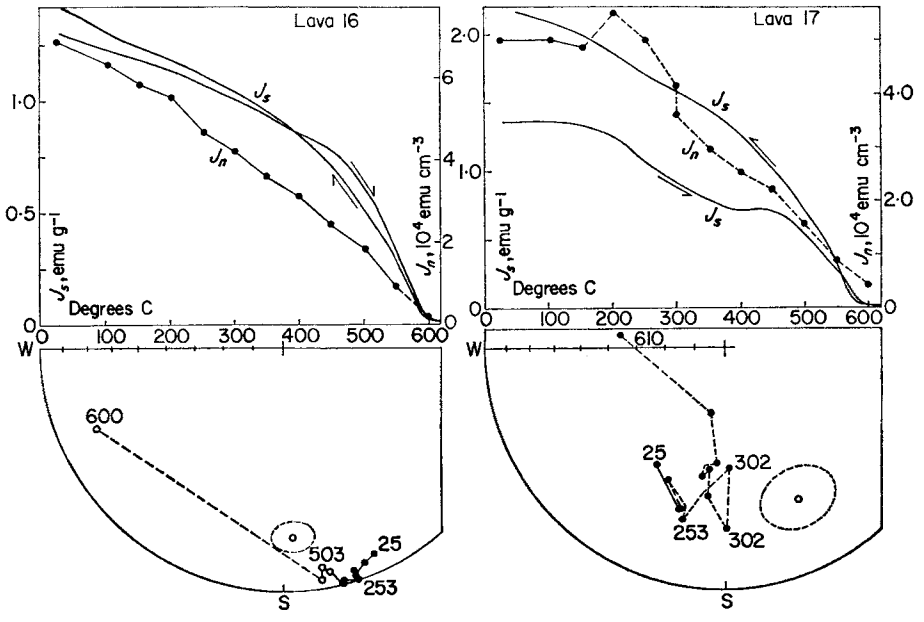


FIG. 6. Thermal demagnetization of NRM and strong-field magnetization, lavas 16 and 17. See Fig. 4 for explanation.

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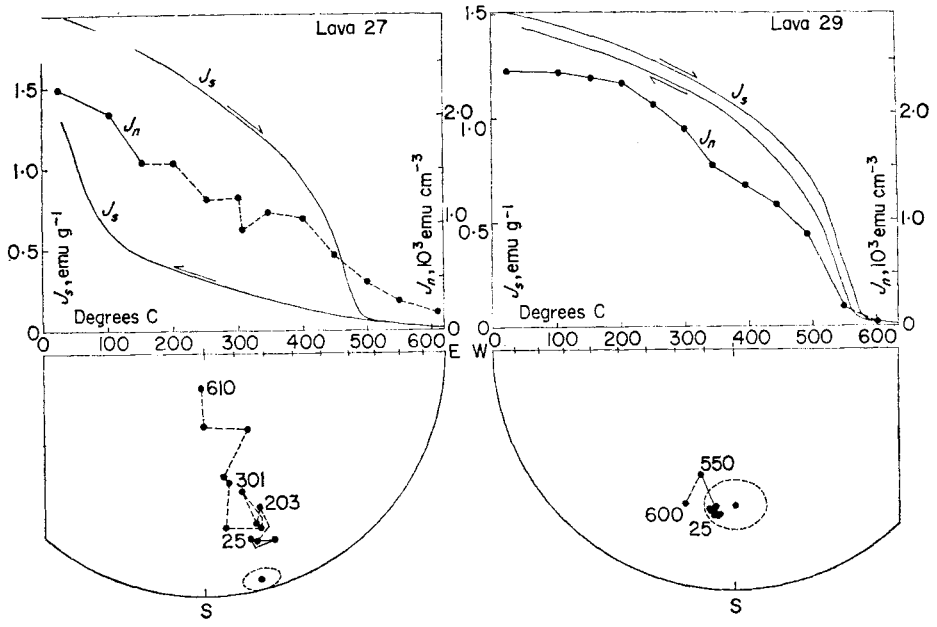


FIG. 7. Thermal demagnetization of NRM and strong-field magnetization, lavas 27 and 29. See Fig. 4 for explanation.

all rocks sampled except lavas 0 and 4. Three distinct types of thermomagnetic behaviour were found, one of which was divided into two subtypes, type 1 and type 1a. Type 1 is illustrated by lava 1 (Fig. 4), lava 17 (Fig. 6), and lava 34 (Fig. 8). The initial Curie temperatures are in the region 250°–350 °C, but a characteristic 'hump' appears in the heating curves around 450°–500 °C, and the cooling curves correspond with material having Curie temperatures from 550°–580 °C and much larger room-temperature magnetizations. This type of thermomagnetic curve has often been observed (e.g., Ozima, Ozima & Kaneoka 1968); it is interpreted as signifying oxidation during heating of initially homogeneous titanomagnetite to a mixture of Ti-poor titanomagnetite and ilmenite (Grommé, Wright & Peck 1969). Type 1a is illustrated by lava 13 (Fig. 4), in which the process of oxidation appears to have been partially completed in nature and more nearly completed during the heating experiment. In type 1a two natural Curie temperatures are found, and the cooling curves are only slightly above the heating curves. In type 1a, as well as in type 1, the NRM appears to reside in the low-Curie-temperature material.

Type 2 magnetization curves are illustrated by lavas 10 and 11 (Fig. 5), lava 16 (Fig. 6), and lava 29 (Fig. 7). Here the cooling curves approximately retrace the heating curves but are below the heating curves for most or all of their lengths. Usually only one Curie temperature is evident, in the vicinity of 570°–585 °C, except in the case of lava 10, where two Curie temperatures are present (Fig. 5). In the lavas giving type 2 curves, the NRM appears to be distributed with respect to temperature much as the strong-field magnetization. The type 2 curves are interpreted as indicating that the lavas were oxidized during cooling; the magnetic material is very likely Ti-poor titanomagnetite containing ilmenite lamellae (Grommé *et al.* 1969).

Only one lava, 27 (Fig. 7), gave a type 3 thermomagnetic curve. The heating and cooling curves are widely dissimilar: the Curie point for the heating curve is 490 °C, that for the cooling curve 100°–150 °C. (Two specimens from this lava flow were

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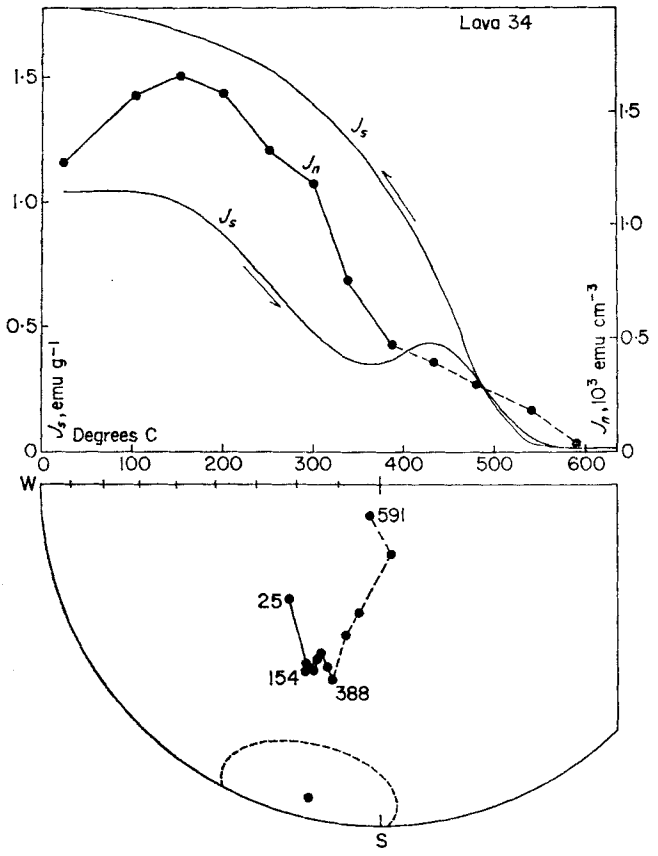


FIG. 8. Thermal demagnetization of NRM and strong-field magnetization, lava 34. See Fig. 4 for explanation.

thermomagnetically analysed; one from the earlier collection was heated in nitrogen, and one from the later collection was heated in air. They gave the same results.) No thermomagnetic curves of this type appear to have been reported previously, and no explanation for the irreversibility is offered here. Note, however, that nearly all of the stable NRM is destroyed upon heating above 150 °C (Fig. 7).

It was noted that lavas 1 and 27 have apparent 'self-reversing' tendencies. Although these two lavas are of very different types, they have in common that two ferromagnetic materials are present over the temperature range in which the tendency toward reversal occurs, and in each case one of the two materials was evidently artificially produced during the laboratory heating. None of the lavas on which thermal demagnetization of NRM was done showed any likelihood of self-reversal in nature.

Correlation of properties

As pointed out above, the grouping of NRM directions may be interpreted to mean that the lavas were erupted in groups with respect to time. In view of the diversity of rock types and thermomagnetic characteristics, it seems worthwhile to examine possible correlations of these parameters with each other and with the

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presumably discrete intervals of time represented by the groups of NRM directions. For this purpose, the thermomagnetic type, petrographic characteristics observable in hand specimen, and the NRM groupings are shown in Table 1.

A fair correspondence exists between the NRM groupings and thermomagnetic types: groups A, B, and E are mostly types 1 and 1a (unoxidized or moderately oxidized), whereas groups C and D are mostly type 2 (oxidized). On the other hand, even though rocks having similar megascopic petrography occur more or less in stratigraphic groups, these petrographic groups do not correspond in any obvious way with the magnetically defined groups. For example, NRM group E contains fine-grained trachytes with and without xenolithic material, olivine basalts, and oxidized porphyritic trachytes. It is possible, of course, that the easily observable characteristics we have used do not sufficiently describe the rocks and that more fundamental parameters such as chemical composition or detailed mineralogy might show correlations with the magnetic groups. Nevertheless, it appears significant that almost all the lavas that appear oxidized (i.e. strongly reddened) in hand specimen are vesicular, and that all of these visibly oxidized rocks give type 1 thermomagnetic curves. When examined more closely, the reddening in these specimens appears confined to vesicle linings and to feldspar phenocrysts; the groundmass remains gray, even in the most reddened samples.

With respect to the geomagnetic reversal, there is no clear-cut lithologic transition. Although the lavas with normal NRM give type 1 thermomagnetic curves, many of the reversed lavas do also. Most, but by no means all, of the visibly oxidized rocks also have normal NRM. It is clear from Table 1 that all parameters such as thermomagnetic or petrographic type, as well as NRM direction, must be considered with respect to time as the independent variable. No casual relations may be inferred among these that are independent of time (*cf.* Wilson & Watkins 1967 for a contrasting viewpoint).

Palaeomagnetic pole and NRM dispersion

Palaeomagnetic poles have been calculated as the means of virtual poles derived from mean NRM directions after a.f. demagnetization (1) from all sites, (2) from sites with reversed NRM, and (3) from sites with normal NRM. The poles and associated dispersion statistics are listed in Table 2. The dispersion of virtual poles

Table 2

Mean palaeomagnetic poles

	N	R	k	δ	α_{95}	N. lat.	W. long.
All sites	22	19·6420	8·91	26·5	11·0	79·5	60·1
Reversed sites	17	15·6964	12·27	22·4	10·6	79·8	57·9
Normal and intermediate sites	5	3·9486	3·80	37·2	45·3	(78·2)	(67·8)

Explanation: δ is the r.m.s. angular dispersion (Wilson 1959); other symbols as in Table 1.

is great; comparison of the dispersion for all sites ($k = 8{\cdot}91$) with that for the reversely magnetized sites only ($k = 12{\cdot}27$) shows that the inclusion for the intermediately magnetized sites (lavas 3 and 4) in the normal group does not unduly increase the overall dispersion. Values of k given in Table 2 are for k_T, expressing the total or observed dispersion. Because the individual site-dispersions are rather great, it is

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desirable to calculate k_B, expressing the between-site dispersion without this contribution to the observed dispersion. The applicable formula is

$$\frac{1}{k_B} = \frac{1}{k_T} - \frac{1}{\bar{N}k_W},$$

where $\bar{N}$ is the mean number of specimens at the various sites and k_W describes the mean dispersion of virtual poles calculated from individual specimens at each site. k_W was obtained from the values of k in Table 1 using the methods of Cox (1970). For $k_T = 8{\cdot}91$, $k_W = 9{\cdot}50$, and $\bar{N} = 5{\cdot}09$, we find $k_B = 10{\cdot}9$, which is not significantly different from k_T. Thus the within-site dispersion makes no great contribution to the total. The high dispersion of the virtual poles (i.e. of the mean NRM directions) is therefore intrinsic to this section of lavas and is not due to the fortuitous inclusion of transient field directions that existed during a geomagnetic reversal.

Direct comparison of the large NRM dispersion in these lavas with other published results is difficult. Cox & Doell (1964) have shown the variation with latitude of dispersion of virtual poles obtained from volcanic rocks younger than 700,000 years at four Western Hemisphere localities. The angular dispersion for rocks from the Galapagos Islands, which are at nearly the same latitude as Ngorongoro, is about 12°, roughly half the value for Ngorongoro. On the other hand, angular dispersion reported by Cox & Doell for Alaskan rocks is nearly 25°. Cox & Doell (1964) also show maps illustrating in angular terms the relative worldwide magnitude of the present non-dipole field. As they point out, their low-latitude localities (Hawaii and the Galapagos Islands) are in an area (the Pacific basin) in which the nondipole field is at present very small. Their maps indicate a rather large non-dipole field at this time over all of the African Continent. In the general area of north and east Africa, the average angular deviation, expressed in terms of virtual poles, between the present field and the inclined dipole field is approximately 12°, in contrast to 3° or less over most of the Pacific basin. It may be possible, therefore, to interpret the large NRM dispersion found in the Ngorongoro lavas as indicating that the large magnitude of the nondipole field now present in the general area of Africa has been a persistent feature.

The mean poles for the normal and reversed lavas are nearly coincident, and the overall mean pole is displaced 10·5° from the geographic pole (Table 2). The 95-per-cent-confidence oval α_{95} for the overall mean pole just includes the geographic pole. Inspection of Fig. 2 suggests, however, that the NRM directions do not approximate Fisher's (1953) distribution owing to the grouping mentioned above; the same is true for the virtual poles, and hence one cannot attach literal significance to the values of α_{95}. (For the same reason, confidence limits (Cox 1969a) have not been placed on the values of k and δ in Table 2.) The apparent lack of conformity to Fisher's distribution may be due to non-uniform or incomplete sampling with respect to the characteristic period of secular variation. For this reason, even though normal and reversed mean poles coincide, no inferences concerning persistent offset of the dipole field seem warranted from these data.

K–Ar ages

Potassium–argon ages were determined on rocks from two locations in the stratigraphic section, using the methods described by Lanphere & Dalrymple (1965), and the results are given in Table 3. All samples in the preliminary collection were examined in thin section, only lava 20 appeared unequivocally suitable for a whole-rock age determination. Argon analyses were made on three specimens cut from the

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Table 3

Potassium–argon analytical data. See text for details of samples 2, 3 *and* 4

	% K_2O	Radiogenic ^{40}Ar, 10^{-12} moles gm^{-1}	Atmospheric ^{40}Ar, % of total ^{40}Ar	Calculated age, 10^6 y
1. Trachyte, whole rock		8·85	85	2·44±0·20
Lava 20	2·46	8·37	90	2·31±0·30
		9·48	76	2·61±0·17
Mean age of lava 20				2·45±0·15
2. Plagioclase		2·40	54	2·23±0·10
	0·73	2·46	81	2·28±0·14
3. Plagioclase with some		3·18	70	2·45±0·15
adherent quartz	0·88	2·92	52	2·25±0·15
4. Plagioclase with 30 per cent adherent quartz	0·85	3·54	78	2·81±0·11

same sample, the three values agree within experimental error, and the best estimate of the age is given by the mean, 2·45±0·15 (Table 3).

Several samples were collected from a non-welded vitric air-fall tuff stratigraphically between lava 0 and lava 1. Plagioclase separated from this material comprises samples 2, 3 and 4 in Table 3. Samples 2 and 3 are different density fractions from the same hand specimen, and sample 4 is from a second hand specimen. The calculated ages differ by more than can be attributed to experimental error, and the reasons are not immediately apparent. The most nearly pure plagioclase concentrate has the lowest K_2O content, the opposite from what one would predict if quartz is the only impurity. This may indicate real differences in plagioclase composition in the different concentrates, raising in turn the possibility of differential argon loss by diffusion from the different plagioclases. On the other hand, the possibility of contamination by older materials derived from the Precambrian basement is always present for these rocks, although they were carefully examined and no signs of contamination were found. The best interpretation of the plagioclase ages is that the extremes of the range of 2·81–2·23 My represent maxima and minima and hence that the tuff between lavas 0 and 1 is not detectably different in age from overlying lava 20.

The age of the transition from normal to reversed geomagnetic polarity is thus dated at 2·45 million years. Cox (1969b) by combining data from oceanic magnetic profiles, ocean sediment cores, and dated volcanic rocks, has placed the age of the transition from the Gauss normal epoch to the Matuyama reversed epoch at 2·43 My our results correspond closely with this.

Summary and conclusions

Of 22 volcanic units sampled in the south-west wall of Ngorongoro, the lowest three are normally magnetized, the next two have intermediate directions, and the upper 17 are reversely magnetized. Potassium–argon dating places this reversal at 2·45 My, corresponding to the boundary between the Gauss normal epoch and the Matuyama reversed epoch. Hitherto this boundary had not been found in a single stratigraphic sequence. The age of the reversal corresponds closely to that given by Cox (1969b), which was based on a synthesis of all available evidence, including oceanic magnetic profiles.

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All of the volcanic rocks were found to have large lightning-produced secondary magnetizations; a.f. demagnetization was moderately successful in preferentially removing these, but partial thermal demagnetization was not. For this reason, no attempts at determining palaeointensities from these rocks were made, but it is possible that such attempts might be successful if a.f. cleaning were combined with stepwise heating according to the procedures of Doell & Smith (1969).

Strong-field thermomagnetic curves obtained from these rocks fall into two main categories: Curie temperatures within the range 250°–350 °C, and Curie temperatures within the range 570°–585 °C. Lavas having the former are presumed to contain unoxidized titanomagnetite, whereas lavas with high Curie temperatures apparently contain titanomagnetite that was oxidized during original cooling.

It was found that the lavas could be divided into five stratigraphically continuous, nonoverlapping groups on the basis of obvious grouping of NRM directions. Such groupings imply that the lavas were not erupted uniformly in time, but rather in groups separated by periods of non-eruption. No simple correlations exist, however, between these NRM groups and such properties as thermomagnetic type or gross petrographic character.

The root-mean-square dispersion of virtual geomagnetic poles corresponding to the mean NRM directions of the 22 volcanic units is 26·5°. This value is high compared with late Quaternary lavas from low latitudes in the Pacific basin (Hawaii and the Galapagos Islands, Cox & Doell 1964). Considered in the light of the fact that the present nondipole field over the African Continent is relatively much greater than that over the Pacific basin, this high dispersion seems less anomalous and might be interpreted to mean that the non-dipole field has been a persistent feature associated with the African Continent during Pleistocene time. The mean palaeomagnetic pole is displaced 10·5° from the geographic pole, but the radius of the 95-per-cent-confidence cone α_{95} is 11·0°. The distribution of virtual poles is apparently non-Fisherian and therefore no exact significance can be attached to the calculated value of α_{95}, but because of the large dispersion and small number of samples, the deviation of the mean palaeomagnetic pole from the geographic pole is probably not significant.

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